

Frontend Specification Document

PrepFlow — Hybrid AI Exam Practice & Progress Tracker

1. Design direction

PrepFlow should feel calm, focused, academic, and trustworthy. It prioritizes readable questions, quick option selection, immediate feedback, useful metrics, and low distraction on mobile.

2. Brand personality

Quality	Interface expression
Focused	Clean layouts and limited decoration
Encouraging	Supportive result feedback without gamified noise
Trustworthy	Source references and clear status
Calm	Soft neutral surfaces and strong readability
Analytical	Simple actionable charts and plain-language insights

3. Color palette

Token	Hex	Use
Primary 700	#1D4ED8	Primary buttons, active navigation
Primary 600	#2563EB	Hover/highlight
Primary 100	#DBEAFE	Selected states
Secondary 700	#0F766E	Secondary emphasis
Success 700 / 100	#15803D / #DCFCE7	Correct/completed states
Warning 700 / 100	#B45309 / #FEF3C7	Revision due/caution
Error 700 / 100	#B91C1C / #FEE2E2	Incorrect/error
Slate 950	#0F172A	Headings
Slate 700	#334155	Body text
Slate 50	#F8FAFC	App background
White	#FFFFFF	Cards/surfaces

4. Typography

Use Inter with system fallbacks. Use Noto Sans/Noto Sans Devanagari where Hindi support is enabled. Questions use at least 16 px on mobile. Use sentence case, high contrast, and readable line heights.

Style	Size	Use
Display	32 px	Desktop hero/key empty state
H1	28 px	Page title
H2	22 px	Section title
H3	18 px	Card/question heading
Body large	16 px	Question text
Body	14 px	General UI

Style	Size	Use
Label	13 px	Inputs/tabs
Caption	12 px	Source/helper text
Metric	24 px	Dashboard KPIs

5. Spacing and layout

Use a 4 px spacing scale: 4, 8, 12, 16, 20, 24, 32, 40, 48 px. Mobile gutter is 16 px; desktop gutter 24–32 px. Practice content max width is 760 px. Minimum touch target is 44 × 44 px.

6. Responsive navigation

Mobile: fixed bottom nav: Home, Practice, Revision, Books, Profile. **Desktop:** left sidebar: Dashboard, Practice, Revision, Books, Analytics, Settings. Show admin navigation only for administrator role.

7. Component styles

Component	Specification
Primary button	Blue #1D4ED8, white 14 px semibold text, 10 px radius, 44–48 px height, visible focus ring, disabled/loading states.
Secondary button	White, slate border, slate text, 10 px radius; used for Back, Cancel, Details.
Option card	Full-card click target; default white/slate border; selected blue; correct green; incorrect selected red; letters A-D in distinct badges.
Input	44 px high, visible label, 8 px radius, blue focus ring, clear error helper text.
Card	White, 14 px radius, subtle border/shadow, 16 px mobile and 20–24 px desktop padding.
Modal	Focus-trapped, Escape closable, confirmation for destructive actions.
Status badge	Clear label plus color: processing, ready, failed, revision due, theory, question bank, mixed.

8. Core screens

- Authentication: centered accessible form card with safe validation.
- Onboarding: exam goal, date, daily target, upload first book, processing status.
- Dashboard: daily progress, Continue Practice CTA, yesterday summary, revision due, weekly trend, weak topics, book progress.
- Books: title, type, processing status, question count, active indicator, set active/view/delete actions.
- Practice: question number/progress, chapter/difficulty, prominent question, A-D option cards, unsure and submit actions.
- Result state: correct/incorrect, correct option, explanation, source page, revision confirmation, Next Question CTA.
- Revision: due now/upcoming/mastered lists and progress.
- Analytics: question volume, accuracy, subject/chapter, revision trend; each chart has text summary.
- Settings: profile, exam, target, timezone, exports, privacy, deletion.

9. Integration and API UX

Supabase Auth handles signup/login/reset. Supabase database stores user-owned records under RLS. Supabase Storage holds original private books via signed URLs. OpenAI is called only by the secure backend for classification, extraction, retrieval, question generation, hints, and explanations.

10. Key application endpoints

Method	Route	Purpose
POST	/api/books/upload	Validate upload and create processing job
GET	/api/books/:bookId/processing-status	Show status
DELETE	/api/books/:bookId	Start secure deletion
GET	/api/practice/next	Return next question without key
POST	/api/practice/submit	Score/save and return feedback
POST	/api/practice/skip	Record skip
GET	/api/revision/due	Return due reviews
GET	/api/analytics/overview	Dashboard data
GET	/api/analytics/export	Secure export

11. Accessibility rules

- Meet WCAG AA contrast for text/actions.
- Never rely only on color for correct/incorrect states.
- Keyboard access for options, submit, navigation, modals, and forms.
- Visible focus states and semantic labels.
- Charts require plain-text summaries and meaningful tooltips.
- Test practice layout at 320 px mobile width and standard desktop widths.